

Coding & Solution

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	5 Marks

Coding & Solution

This section presents the implemented code and the overall solution for FitBuddy.

Solution Summary

Repository Link / URL	https://github.com/Dhanush-05-AI/fitbuddy-1
Programming Language(s)	Python 3.12, HTML, CSS, JavaScript
Framework(s) Used	FastAPI, Uvicorn, Jinja2, SQLAlchemy, Pydantic, google-genai (Google Gemini API)
Key Features Implemented	1. Input form with validation 2. AI 7-day workout plan (Gemini) 3. Nutrition / recovery tip 4. Feedback-based plan update 5. Database storage of users and plans 6. Admin page to view and delete users 7. Retry, backoff and friendly error pages 8. Health check and Docker deployment
Pending / Incomplete Features	Password protection must be enforced on the admin routes (/view-all-users, /delete-user) User accounts, progress tracking and cloud deployment (future work)
Setup / Run Instructions	1. git clone https://github.com/Dhanush-05-AI/fitbuddy-1.git 2. Create and activate a virtual environment 3. pip install -r requirements.txt 4. Copy .env.example to .env and set GOOGLE_API_KEY, ADMIN_USERNAME, ADMIN_PASSWORD 5. python run.py 6. Open http://127.0.0.1:8000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes

S.No	Criteria	Status (Yes / No)
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Team: Dhanush M (Team Leader), Jayavardhani P, Sandhiya S, V Brindha, Ishwarya. Team ID: SWTID-2026-5841.

Automated tests are in tests/test_app.py (pages, generate-and-feedback flow, validation, admin access, Gemini failure handling).

Secrets are kept in .env, which is excluded from the repository; .env.example shows the required variables.